

MSSE Capstone Project

Sprint Planning & Project Initiation Meeting Minutes

TrustAI Marketplace

Date	July 8, 2026
Time	7:00 PM CAT
Location	Virtual meeting
Meeting Type	Product vision review, Sprint 0 planning, architecture and delivery alignment
Prepared For	TrustAI Marketplace MSSE Capstone team records

1. Attendees

Five active team members attended and participated: Ahmed Al-Mandalawi, Ranga Nyamadzawo, Mulima Chibuye, Adrian Muchatibaya, and Samar El Ghandour.

The team confirmed that it would proceed with five active members. Abdallah Mohmoud, who had been listed at the July 4 kickoff, had not participated or confirmed continued involvement. The team agreed not to delay the project, and responsibilities associated with that workstream would be redistributed across Ahmed Al-Mandalawi, Ranga Nyamadzawo, Mulima Chibuye, Adrian Muchatibaya, and Samar El Ghandour.

2. Meeting Objective

The meeting was held to convert the selected project idea into an executable delivery plan. The immediate goal was to confirm the product vision and MVP boundaries, review the proposed sprint schedule and backlog, agree on the technical direction, identify risks, and make sure work could begin immediately.

- Confirm the product vision and define what is inside and outside the MVP.
- Review the four-sprint delivery plan and preserve a final review buffer.
- Confirm active team roles and redistribute work after the team reduction.
- Agree on the repository, Trello board, documentation, CI/CD, Docker, and deployment approach.
- Discuss the proposed technology stack and AI/LLM strategy.
- Resolve how pricing or market-value information should be represented in the MVP.
- Identify architecture, wireframing, advisor check-in, and immediate action items.

3. Administrative and Capstone Check-In Matters

The team discussed whether it needed to formally notify Quantic that the group had been formed or submit the project idea in advance. The working understanding was that no separate up-front

declaration had been requested. The team will complete the required group contract and submit it with the capstone materials when required.

The team also reviewed Quantic's capstone check-in week and the office hours offered by Dr. Robert Still. Members agreed that an advisor check-in would be useful, but that the meeting would be more productive after the architecture, backlog, and main questions had been prepared.

Advisor Check-In Direction

Continue implementation now rather than waiting for the advisor meeting. Prepare the architecture and refined backlog over the weekend, hold a short internal alignment check if needed, and then attend the advisor session with specific questions about scope, architecture, and rubric alignment.

4. Product Vision

Ahmed Al-Mandalawi restated the product vision for TrustAI Marketplace. The product is an AI-assisted marketplace decision-support application intended to make online purchasing decisions safer, easier, and more informed.

A user will submit a marketplace listing as pasted text, structured information, or a URL. The application will analyze the available details and return a clear decision-support report. The report is expected to include scam indicators, a risk score, a price-plausibility assessment, a plain-English recommendation, and suggested questions the buyer should ask before completing the purchase.

Product Vision

Help online marketplace buyers evaluate a listing before they buy by turning incomplete or suspicious listing information into an explainable risk assessment and practical next steps.

5. MVP Scope

The group agreed that the MVP must emphasize core functionality and capstone learning outcomes rather than attempting to build a fully commercial marketplace platform.

5.1 In Scope for the MVP

- User registration, login, secure sessions, and user profile support.
- Submission of a listing through pasted text, a URL, or structured listing details.
- Storage and processing of submitted listings.
- AI-assisted analysis for scam indicators, risk scoring, recommendation, and buyer questions.
- A price-plausibility result that explains whether the asking price appears plausible, suspicious, or unusually low.
- A React interface that displays the analysis clearly, including loading, validation, and error states.
- User-specific analysis history, which is one reason authentication was retained.
- Automated testing, Docker-based development, CI/CD, documentation, and a live deployable demonstration.

5.2 Outside the MVP / Stretch Work

- Image-based listing analysis unless all core features are complete and stable.
- A browser extension, which was considered a promising post-capstone enhancement.
- Direct integrations with multiple marketplaces.
- A production-grade factual market-value engine based on large-scale scraping.
- Mobile applications, payments, or commercial subscription features.

Authentication was viewed as a relatively straightforward feature and not the main differentiator of the product. The team nevertheless agreed that it is useful because it demonstrates secure application design and enables saved analysis history. The distinctive parts of the project - listing analysis, explainable risk, and price plausibility - should receive more attention.

6. Delivery Timeline and Sprint Cadence

Mulima presented a schedule running through August 31, 2026. The team discussed whether four full two-week sprints could fit while preserving a final two-week buffer. After checking the dates, the group agreed that Sprint 0 should be short, followed by three main development sprints and a final review/submission week.

Agreed Schedule

Sprint 0: July 7-12 (setup and planning). Sprint 1: July 13-26 (foundations). Sprint 2: July 27-August 9 (core end-to-end value). Sprint 3: August 10-23 (complete and harden the MVP). Final review week: August 24-31 (regression testing, documentation, presentation, and submission).

The agreed sprint plan was for each sprint to finish with something demonstrable. The team did not want separate components to remain unintegrated until the end. At this planning stage, the intended sprint-boundary evidence was a working increment, an application demo recording, and updated documentation.

7. Active Team Roles and Work Redistribution

The main leadership roles remained unchanged: Ahmed Al-Mandalawi continued as Product Owner and AI Analysis Lead, while Mulima Chibuye continued as Project Manager / Scrum Master. With five active members, the listing/data and pricing responsibilities associated with the departed workstream had to be redistributed.

Mulima Chibuye - Project Manager / Scrum Master

Trello, sprint planning, risk and dependency tracking, meeting facilitation, status reporting, documentation coordination, and advisor check-in preparation.

Ahmed Al-Mandalawi - Product Owner / AI Analysis Lead

Product vision, priorities, acceptance criteria, architecture overview, repository handoff, LLM abstraction, scam/risk analysis, recommendations, and price-plausibility model.

Ranga Nyamadzawo - Backend Lead - Auth, Listings & Data

Registration/login/JWT, listing ingestion and processing, database schema and migrations, pytest foundations, and data support for price plausibility.

Adrian Muchatibaya - Frontend Lead

Wireframes and mockups, React authentication screens, listing submission, results, history, responsive UI, and API integration.

Samar El Ghandour - QA & DevOps Lead

Test strategy, Docker environments, GitHub Actions, deployment workflow, release quality gates, and end-to-end validation.

Ownership means accountability, not isolation. Pairing is expected where services meet, and members may pull additional cards after completing their assigned work. Trello will be used to make ownership visible and prevent two people from unknowingly working on the same task.

8. Sprint 0 Priorities

Sprint 0 is intentionally short and focused on removing uncertainty so the team can enter Sprint 1 with a shared foundation. The meeting confirmed the following Sprint 0 priorities:

- Create and populate the Trello Scrum board; collect member email addresses and invite all active members.
- Adopt and share Ahmed's existing skeletal GitHub repository with the team and the Quantic grader.
- Define the repository structure, branch strategy, pull-request review rules, and documentation folders.
- Produce an end-to-end architecture diagram and bounded-context view.
- Draft the Postgres schema for users, listings, and analyses.
- Create wireframes and mockups for authentication, submission, results, and history flows.
- Set up Docker/docker-compose so the project behaves consistently across Windows, macOS, and Linux/WSL.
- Create the initial GitHub Actions workflow for linting and pytest.
- Make the frontend skeleton available at the custom-domain deployment.
- Complete an LLM/provider spike and define deterministic mock responses for development and CI.
- Refine, estimate, and assign the backlog before Sprint 1 begins.

9. Repository, Documentation, and Work Tracking

The team agreed to use Ahmed Al-Mandalawi's existing skeletal repository rather than begin again with an empty project. Ahmed confirmed that it could be shared with the group and developed collaboratively through branches and pull requests.

Project documents will also be maintained in the repository as Markdown under a dedicated /docs area. This will include meeting minutes, the project-management history, architecture decisions,

diagrams, sprint reports, and the decision log. Version-controlling the documents will make changes visible and support the final capstone submission.

Mulima will maintain Trello as the operational source of truth. Cards will move through Backlog, Sprint Backlog, In Progress, In Review, and Done. Members should assign a card to themselves before starting work. If a member finishes early, they may pull another unassigned item as long as the sprint priorities remain protected.

10. Development Environment, CI/CD, and Deployment

The team compared operating systems and confirmed a mix of Windows, macOS, Linux, and Windows Subsystem for Linux. Docker was therefore accepted as an important part of the technical scope because it reduces host-specific setup differences.

The team agreed that DevOps work should run in parallel with feature development. Linting, unit tests, and pull-request checks should be added early rather than postponed. A CI/CD pipeline should evolve throughout the project, with improvements made as the services and test suites mature.

The existing custom-domain site uses GitHub Pages. The frontend can be deployed there from Sprint 0. The team must still confirm the hosting arrangement for FastAPI services and Postgres, because GitHub Pages hosts static content and cannot by itself run the backend or database.

Deployment Clarification Required

Keep the custom domain and GitHub Pages for the frontend. During Sprint 0, select and document a compatible backend/database hosting option, then connect it to the same CI/CD process. This prevents the team from discovering a hosting limitation late.

11. Proposed Technology Stack

The meeting produced a practical, mostly free or low-cost stack that builds on tools already used in the MSSE program.

- Backend: Python with FastAPI, organized into clear services or bounded contexts.
- Frontend: React.
- Database: PostgreSQL, with SQLAlchemy and a migration tool; SQLite may be used only for limited local experimentation if needed.
- Testing: pytest for backend business logic and service tests, followed by integration and end-to-end tests.
- Containers: Docker and docker-compose for repeatable local environments.
- CI/CD: GitHub Actions, with linting and tests on pull requests and deployment from the main branch.
- AI: provider-switchable LLM integration; Groq was favored for low cost, with other providers kept replaceable through configuration.
- Development/CI AI mode: deterministic mock responses so tests remain repeatable and do not consume API credits.

The team discussed the possibility of making the AI provider configurable so that the application can switch providers without rewriting business logic. Benchmarking different providers for quality, latency, and cost was also suggested.

12. Architecture and Design Deliverables

Members requested an end-to-end architecture view before full development begins. The diagram should show the React frontend, backend services, database, AI provider interface, and the main request flow. This will help the team understand where each workstream fits and will guide backlog priorities and API contracts.

Wireframes and mockups were also added as a formal deliverable. The team needs a shared visual flow for authentication, listing submission, analysis results, and history before the frontend is built in detail. The group emphasized that these artifacts are not decorative; they provide evidence for why the chosen design was implemented and support the final project defense.

13. Pricing and Market-Value Discussion

A major open question was how the application would calculate market value if direct marketplace integrations and large-scale scraping are outside the MVP. Members correctly noted that an LLM should not be allowed to invent a precise numerical market value without a verifiable data source.

Potential data sources such as eBay, Facebook Marketplace, Craigslist, Alibaba, retailers, OEM sites, and currency services were discussed. The group also noted the practical problems: bot blocking, limited API access, differences between new and used items, regional pricing, currency conversion, and the risk of hallucinated values.

MVP Pricing Decision

For the MVP, replace a factual numerical market-value claim with a price-plausibility assessment. The system may classify the asking price as plausible, suspicious, or too good to be true and explain the risk signals. A verified comparable-price engine can remain a later enhancement if a reliable API or curated dataset is found.

This approach still supports the product's safety goal while remaining explainable and realistic. The AI can interpret listing evidence and provide a narrative, but the application should avoid presenting unsupported values as fact.

The group briefly discussed whether fetched information would make the system a RAG application. The working view was that simple retrieval or fetched context supplied with a prompt may be better described as in-context enrichment unless a persistent retrieval knowledge base is deliberately implemented.

14. Risks and Open Questions

- Reduced capacity: five active members must absorb the departed member's responsibilities without expanding the MVP.
- Schedule pressure: Sprint 0 is very short, so setup tasks must be completed immediately.
- Backend hosting: the static GitHub Pages deployment does not host FastAPI or Postgres; a backend platform must be selected.
- AI cost and rate limits: use free tiers, provider switching, caching, and deterministic mocks.
- AI quality: combine clear rules and structured prompts; define measurable acceptance criteria.
- Pricing reliability: do not claim precise market value without trustworthy comparables.

- Marketplace access: URLs may be blocked or unreachable; errors must be handled transparently.
- Cross-platform setup: Docker must be validated on Windows and Mac early.
- Integration risk: agree service contracts and integrate every sprint.
- Documentation risk: update /docs continuously rather than reconstructing evidence at the end.
- Timezone and response delays: use asynchronous updates and direct contact for urgent blockers.
- Scope creep: image analysis and browser extensions stay out of the MVP unless all Must items are secure.

15. Decisions Made

- Proceed with five active team members and redistribute the departed member's tasks.
- Retain user registration and login so analyses can be associated with user history.
- Use a short Sprint 0, three main build sprints, and a final review/submission week ending August 31.
- Begin work immediately; do not wait for the advisor check-in.
- Adopt Ahmed's existing skeletal repository and use pull requests for collaboration.
- Maintain project documentation as Markdown under /docs in the repository.
- Use Trello to assign and track backlog items.
- Use FastAPI, React, PostgreSQL, Docker, pytest, and GitHub Actions as the working technical direction.
- Design the LLM integration so providers can be switched through configuration; use deterministic mocks in development and CI.
- Use price plausibility rather than an unsupported factual market-value estimate in the MVP.
- Create architecture diagrams and wireframes during Sprint 0.
- Deploy the frontend early on the custom domain and confirm separate backend/database hosting.

16. Action Items

The following actions consolidate what was agreed in the meeting and the immediate Sprint 0 follow-up reflected in the updated project plan and backlog.

Owner	Action	Target	Status
Ahmed	Share the existing skeletal repository; add the /docs structure; prepare the end-to-end architecture and bounded-context diagrams; refine product priorities and price-plausibility acceptance criteria.	Sprint 0 / immediate	Open
Mulima	Update the project plan and backlog for five members; redistribute tasks; create/populate Trello; collect emails; prepare the advisor check-in agenda and schedule.	Sprint 0 / immediate	In progress
Ranga	Begin backend foundations for authentication, listings, database schema/migrations, pytest, and support the price-plausibility data approach.	Sprint 0-1	Open
Adrian	Produce wireframes and mockups for auth, listing submission, analysis results, and history; prepare the React skeleton.	Sprint 0	Open
Samar	Draft the QA/DevOps approach; validate Docker across team operating systems; create the CI skeleton; evaluate frontend and backend deployment workflows.	Sprint 0	Open
All members	Share the email used for Trello/GitHub; join the board and repository; review the updated backlog; select assigned cards; begin work immediately and raise blockers in chat.	Immediately	Open

Owner	Action	Target	Status
Team	Confirm backend and Postgres hosting; validate the LLM/provider shortlist; agree initial API contracts; prepare focused questions for the advisor check-in.	Before/at advisor check-in	Open

17. Advisor Check-In Preparation

The team intends to use the Quantic advisor session to validate direction rather than to ask for the project to be designed for them. Before the session, the team should have the architecture, MVP scope, updated backlog, deployment question, and price-plausibility decision ready.

Suggested questions for the advisor:

- Is the revised MVP scope appropriately sized for five part-time team members?
- Does the proposed service decomposition demonstrate sufficient architecture without creating unnecessary complexity?
- Is a price-plausibility assessment an acceptable and more defensible alternative to factual market-value estimation?
- Is the proposed evidence set - backlog, sprint demos, architecture, tests, CI/CD, and documentation - aligned with the rubric?
- Are there any concerns with the planned deployment split between a static frontend and hosted backend/database?

18. Communication and Working Agreement

The team agreed not to wait for weekly meetings to resolve blockers. WhatsApp and Slack will remain the main asynchronous channels, and direct calls are acceptable for urgent issues when a member is not responding. Members acknowledged the timezone differences and agreed to be reachable when awake and to respond as soon as practical.

The working expectation is simple: each member should always have a visible assigned task, update its Trello status, and communicate early when blocked. This is especially important now that the schedule is compressed and the team has reduced capacity.

19. Next Meeting and Immediate Next Step

The team will continue through chat immediately after the meeting. A short internal alignment meeting may be held once the updated backlog, architecture, repository, and wireframes are available. The advisor check-in will then be scheduled with a focused agenda.

The immediate sequence is:

- Share the repository and Trello invitations.
- Publish the revised backlog and work redistribution.
- Assign Sprint 0 cards and begin work.

- Complete architecture, wireframes, Docker/CI setup, schema, and LLM spike.
- Review progress and blockers before the advisor check-in.
- Enter Sprint 1 with agreed contracts, owners, and a demonstrable setup baseline.

20. Closing Notes

The meeting ended with agreement that the planning phase is complete enough for implementation to begin. The team recognized that the project no longer has room for avoidable delays. Members committed to using the board, communicating between meetings, and building in parallel while integrating frequently.

The central delivery principle remains: build the smallest credible version of TrustAI Marketplace, demonstrate the engineering practices taught in the MSSE program, and preserve enough time at the end for testing, documentation, and presentation quality.